

Wesley Rancher

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Summary

Geospatial data analyst focused on remote sensing, machine learning, and data engineering. Experienced in building reproducible workflows for satellite, drone, and enterprise datasets using Python, SQL, and modern ML frameworks. Work centers on segmentation, classification, spatial analytics, and automation across environmental and infrastructure domains.

Education

University of Oregon | M.S. in Geography, GPA: 3.9

Ohio Wesleyan University | B.A. in Environmental Studies and Geography, GPA: 3.5

Skills

Languages: Python, R, SQL, Bash

Machine Learning: PyTorch, Scikit-learn, Random Forest, U-Net, feature engineering

Spatial: ArcGIS Pro/Enterprise, ArcGIS Online, GEE, Planetary Computer, Rasterio, GeoPandas, GDAL

Data Engineering: NumPy, Pandas, Xarray, Docker, Git, REST APIs, AWS CLI

Applications: Remote sensing, change detection, ecological modeling, spatial analysis

Professional Experience

GIS Analyst – American Electric Power

October 2025 – Present

Hydro Asset Computer Vision (Ongoing)

- U-Net segmentation on RGB and NIR drone imagery to identify shoreline infrastructure and support site inspections for a major hydro asset
- Training data generation through manual annotation, spectral indices, thresholding, and Random Forest-assisted labeling
- Preprocessing pipelines using Rasterio and GeoPandas for orthomosaic prep, tiling, mask creation, and inference datasets

Enterprise Data Engineering (Ongoing)

- ETL pipeline development using Python, SQL, REST APIs, IBM Tririga, and ArcGIS to synchronize infrastructure GIS records with enterprise databases
- Schema alignment workflows that standardize geospatial attributes and reduce manual processing across systems

Real Estate Transactions Web App (Ongoing)

- ArcGIS Online application supporting parcel tracking and transaction monitoring across eleven states
- Smartsheet integration using Smartsheet-SDK and ArcGIS Python libraries for centralized feature updates and data management

GIS Automation

- Optical character recognition workflows converting metes-and-bounds descriptions from historical property deeds into draft parcel geometries
- ArcPy automation converting engineering CAD drawings into geospatial features to reduce manual digitization and improve consistency

Graduate Research Assistant – University of Oregon

September 2023 – August 2025

Advisor: Dr. Melissa Lucash

- Collaborated with researchers from the Bonanza Creek Long Term Ecological Research (LTER) program to evaluate wildfire-driven vegetation change and ecosystem dynamics under a changing climate
- Developed an end-to-end remote sensing workflow integrating multispectral time series, Google Earth Engine, Random Forest regression models, and LANDIS-II simulations to quantify aboveground biomass dynamics across eight million hectares of interior Alaska
- Built image preprocessing pipelines for multi-decadal Landsat imagery, including atmospheric and topographic correction, spectral index calculation, and cross-sensor harmonization
- Containerized the LANDIS-II ecosystem model using Docker to improve reproducibility and simplify deployment of large-scale landscape simulations in cloud environments

Graduate Research Fellow – NASA DEVELOP

June 2023 – August 2023

Advisor: Dr. Anthony Vorster

- Developed random forest models integrating vegetation indices, LiDAR, and field observations to map invasive vegetation throughout the Paria River watershed
- Processed multispectral imagery and LiDAR using ArcGIS Pro and Google Earth Engine to generate reproducible geospatial products supporting land management decisions
- Presented technical findings to federal and nonprofit partners while co-authoring the final project report

Undergraduate Research Assistant – Ohio Wesleyan University

December 2022 – May 2023

Advisor: Dr. Nathan Rowley

- Estimated glacial lake depth in western Greenland using radiative transfer methods and Landsat imagery, resulting in a peer-reviewed publication in *Glaciers*
- Refactored a MATLAB-based raster sieving workflow into R, reducing code complexity while simplifying the process of raster-based feature delineation
- Developed reproducible remote sensing workflows for NDWI-based glacial lake mapping and analysis

Undergraduate Research Fellow – University of Central Oklahoma

June 2022 – July 2022

Advisor: Dr. Victor Gonzalez

- Conducted field and laboratory experiments evaluating thermal tolerance of pollinators under environmental stress
- Co-authored peer-reviewed publication in the *Journal of Experimental Biology* and presented at the International Union for the Study of Social Insects

Teaching Experience

Graduate Teaching Assistant – University of Oregon

September 2023 – Spring 2025

Geography 485/585: Remote Sensing I

- Developed lab exercises, taught GIS and remote sensing software (ArcGIS, QGIS, R), and led hands-on demonstrations to undergraduates and graduates

Geography 199: Global Wildfire

- Supported curriculum development, supplemental instruction, student assistance
- **Guest lecture:** “*Changing Wildfire in Brazil*” – discussed landscape drivers of a changing fire regime in Brazil
- **Guest lecture:** “*Bees and Wildfire*” – explained the interplay between post-wildfire effects, vegetation, and pollinators

Awards and Honors

Rippey Research Grant (\$1000, University of Oregon)	2024
NASA Develop Scholarship (\$1500, Science Systems and Applications Inc)	2023
Dean's List (Ohio Wesleyan University)	Fall '22, Spring '20, '22, '23
Robert E. Shanklin Distinguished Scholar (Geography, Ohio Wesleyan University)	2023
Phi Sigma Tau (Philosophy, Ohio Wesleyan University)	2023
Our New Gold Digital Storytelling <u>winner</u> (Spanish, Ohio Wesleyan University)	2022

Publications

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- Weiss, S., Rancher, W., Hayes, K., Buma, B., & Lucash, M. (2026). *Wildfire dynamics under climate change in interior Alaska. (Accepted pending minor revisions)*
 - Lucash, M., Lamping, J., Nowell, B., Scheller, R., Banerjee, T., Buettner, C., Fawcett, J., Hurteau, M., Parks, S., Rancher, W., Robbins, Z., St. Denis, L., Stasiewicz, A., Urza, A., & Weiss, S. (2026). *Roadmap for the future of extreme wildfire events. Fire Ecology*
 - Gonzalez, V., Rancher, W., Vigil, R., Garino-Heisey, I., Oyen, K., Tscheulin, T., Petanidou, T., Hranitz, J., & Barthell, J. (2024). *Bees remain heat tolerant after acute exposure to desiccation and starvation. Journal of Experimental Biology*
 - Rowley, N., Rancher, W., & Karmosky, C. (2024). *Comparison of multiple methods for supraglacial melt-lake volume estimation in western Greenland during the 2021 summer melt season. Glaciers*

Presentations

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- Rancher, W. (2025). *Estimating species-level aboveground carbon in interior Alaska using machine learning and process-based models. University of Oregon, Eugene, OR. (Master's thesis presentation)*
 - Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2025). *Estimating recent shifts in aboveground carbon and species composition in interior Alaska using Landsat imagery and random forests. Northwest Scientific Association, Eugene, OR. (Poster)*
 - Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2024). *Assessing vegetation shifts in boreal Alaska by integrating Landsat imagery with spatial modeling. American Geophysical Union, Washington, DC. (Poster)*
 - Rancher, W., VanArnam, M., Kowalski, A., Anarella, T., & Vorster, A. (2023). *Mapping Russian olive and tamarisk to inform invasive species management along the Paria River, Utah. NASA DEVELOP Day, Washington, DC. (Virtual talk)*
 - Rancher, W., Rowley, N. (2023). *Estimating supraglacial melt lake volume changes in west-central Greenland using multiple remote sensing methods. Ohio Wesleyan Spring Symposium, Delaware, OH. (Poster)*
 - Rancher, W., Vigil, R., Garino-Heisey, I., & Gonzalez, V. (2022). *Effects of desiccation on bees' heat tolerance. Ohio Wesleyan Connection Conference, Delaware, OH. (Poster)*
 - Rancher, W., & Gonzalez, V. (2022). *Effects of desiccation on bees' heat tolerance. IUSSI Sección Andina y del Caribe, Panama City, Panama. (Talk)*